

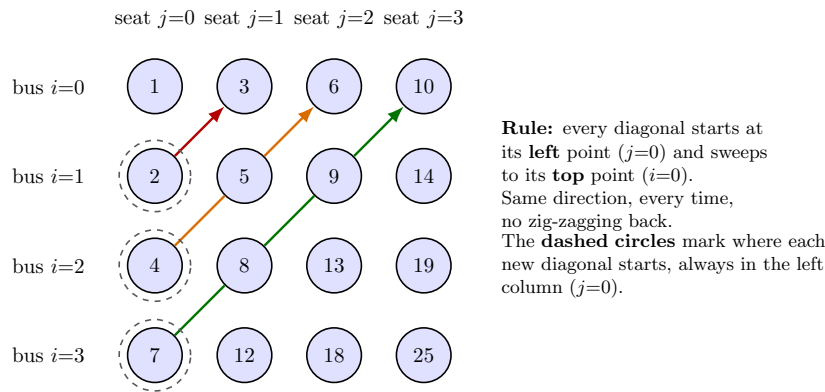
Hilbert's Hotel, Act 3: the diagonal pairing, made rigorous

Setup. Bus i ($i = 0, 1, 2, \dots$), seat j ($j = 0, 1, 2, \dots$). We want ONE room-number function $f : \{(i, j)\} \rightarrow \mathbb{N}$ that hits every room exactly once, so all buses' passengers fit in the hotel.

Intuition (the diagonals). Group pairs by $d = i + j$. Diagonal d is the set of pairs with $i + j = d$: it has exactly $d + 1$ points, running from $(i, j) = (d, 0)$ (its **left** end) to $(0, d)$ (its **top** end). Diagonal 0: just $(0, 0)$. Diagonal 1: $(1, 0), (0, 1)$. Diagonal 2: $(2, 0), (1, 1), (0, 2)$. Etc. Every pair belongs to exactly one diagonal, so if we number diagonal-by-diagonal, sweeping *left to top* within each diagonal, we number every pair exactly once.

Derivation of the formula. Let $T_d = \sum_{k=1}^d k = \frac{d(d+1)}{2}$ (with $T_0 = 0$) be the number of points in *all earlier diagonals* $0, 1, \dots, d-1$ combined (diagonal $k-1$ has k points, for $k = 1, \dots, d$). Within diagonal d , the point (i, j) with $i + j = d$ sits at position $j + 1$ counting from the left end ($j = 0$ is 1st, $j = 1$ is 2nd, $\dots$). So:

$$f(i, j) = \underbrace{T_{i+j}}_{\text{rooms used by earlier diagonals}} + \underbrace{(j+1)}_{\text{position within this diagonal}} = \sum_{k=1}^{i+j} k + (j+1).$$



Why $j + 1$, and not $i + 1$?

The offset is smallest ($= 1$) at $j = 0$ and largest ($= d + 1$) at $j = d$. Since $j = 0$ is the **left** end of the diagonal, $j + 1$ is exactly what makes numbering start at the left and finish at the top. Using $i + 1$ instead would be smallest at $i = 0$ (the **top**), so it would start from the top and finish at the left, the opposite of what we want. **So $j + 1$ is correct as is; no formula change needed** — only the picture needed fixing.

Injectivity. Suppose $f(i, j) = f(i', j')$ with $d = i + j$, $d' = i' + j'$, and say $d \leq d'$. The largest value on diagonal d is $f(0, d) = T_d + (d + 1)$; the smallest value on diagonal $d + 1$ is $f(d + 1, 0) = T_{d+1} + 1$. Their difference is $T_{d+1} - T_d + 1 - (d + 1) = (d + 1) - d = 1 > 0$, so diagonal $d + 1$'s smallest value is exactly 1 more than diagonal d 's largest, meaning *different diagonals never share a value* (their ranges are disjoint and consecutive). So $f(i, j) = f(i', j')$ forces $d = d'$. Then $T_{i+j} = T_{i'+j'}$ cancels, leaving $j + 1 = j' + 1$, so $j = j'$, and then $i = d - j = d' - j' = i'$. Hence $(i, j) = (i', j')$: f is injective.

Surjectivity. Given any $n \in \mathbb{N}$, since T_d is strictly increasing in d and $T_0 = 0$, there is a unique $d \geq 0$ with $T_d < n \leq T_{d+1}$ (the diagonal n falls into). Set $j = n - T_d - 1$ and $i = d - j$. Since $1 \leq n - T_d \leq d + 1$, we get $0 \leq j \leq d$, so $i = d - j \geq 0$ too, i.e. (i, j) is a genuine valid pair. Then $f(i, j) = T_{i+j} + (j + 1) = T_d + (n - T_d - 1) + 1 = n$. So every $n \in \mathbb{N}$ is hit: f is surjective.

Conclusion. f is injective and surjective, hence a **bijection** $\{(i, j) : i, j \geq 0\} \rightarrow \mathbb{N}$. Every bus's every seat gets a distinct room, and every room is used, exactly as in Act 3. This is Theorem 2: $\mathbb{N} \times \mathbb{N}$ (equivalently, any two countably infinite sets' product) is countable.

Hilbert's Hotel only needs an injection (you're right)

I want to flag something that's easy to blur together, because it's exactly why I picked the diagonal formula for the slides instead of the other popular solution you'll see online. The Hotel's actual problem is only: can every guest, old and new, get a room to themselves, with nobody ever forced to share? That's an **injection** from (bus, seat) pairs into room numbers, nothing more. It does *not* require that every room number gets used. Proving Theorem 2, that $\mathbb{N} \times \mathbb{N}$ is exactly the same size as $\mathbb{N}$, is a strictly stronger claim: that needs a full **bijection**, injective *and* surjective. This gap is exactly where the prime-power method, the one behind the TED-Ed and Veritasium tellings of this story, lives.

The solution. Send bus i , seat j (with $i, j \geq 0$) to room

$$g(i, j) = 2^i \cdot 3^j.$$

Proposition. g is injective: distinct bus-seat pairs always land in distinct rooms.

Proof. Suppose $g(i, j) = g(i', j')$, i.e. $2^i 3^j = 2^{i'} 3^{j'}$. By the Fundamental Theorem of Arithmetic, every natural number has exactly one prime factorization, so the exponent of 2 must agree on both sides, giving $i = i'$, and the exponent of 3 must agree, giving $j = j'$. Hence $(i, j) = (i', j')$, so g is injective. ■

That's already enough to solve the Hotel: no matter how many buses show up, or how many seats each has, every single guest gets a private room number that nobody else has been given. Problem solved, in the sense the story actually cares about.

But some rooms are never assigned

g only ever outputs numbers whose sole prime factors are 2 and 3 (numbers like 1, 2, 3, 4, 6, 8, 9, 12, 16, 18, ...). Take room 5: it's prime and isn't 2 or 3, so no (i, j) gives $2^i 3^j = 5$. Same story for 7, and for $10 = 2 \times 5$: that stray factor of 5 can never come from any power of 2 and 3, so 10 is skipped too. Checking every room up to 29: rooms 5, 7, 10, 11, 13, 14, 15, 17, 19, ... all sit permanently empty under this assignment. So g is injective but **not surjective**, hence **not a bijection**.

That's perfectly fine for the Hotel scenario, injective is all it ever needed, but it means this construction alone can't be used to prove Theorem 2, since Theorem 2 is a claim about $\mathbb{N} \times \mathbb{N}$ and $\mathbb{N}$ having exactly the same size, and an injection that skips infinitely many rooms doesn't establish that on its own (you'd need to separately invoke "injects into a countable set $\Rightarrow$ countable" to finish the job). This is exactly why my slides use the diagonal pairing from page 1 instead: since it's a genuine bijection, and a bijection is in particular also an injection, one single construction solves the Hotel's room-assignment problem *and* proves Theorem 2 at the same time. The prime-power trick is a lovely, quick way to see why nobody ever collides, but the diagonal enumeration is the one formula that does both jobs at once.

Two more popular tellings: Veritasium and TED-Ed

Veritasium's "flattening." What he calls flattening the 2D grid into a single line is the same diagonal construction as page 1, just described physically: snake through the grid diagonal by diagonal and lay the diagonals end to end into one list. That list *is* the function $f(i, j)$ from page 1. No new formula here, just a different picture of the same bijection.

TED-Ed's construction. This one is genuinely different, and it's solving a harder version of the story: the hotel is already full (guests sitting in rooms $1, 2, 3, \dots$) *and* infinitely many buses arrive at once, each with infinitely many passengers. You've got the recollection right, here it is made precise.

Since there are infinitely many primes (Euclid's classical theorem, so we can never run out no matter how many buses show up), list them as $2 = p_1 < p_2 < p_3 < \dots$ (so $p_1 = 2, p_2 = 3, p_3 = 5, p_4 = 7, \dots$). Reserve $p_1 = 2$ for the old guests, and hand bus m ($m = 0, 1, 2, \dots$) the prime $q_m := p_{m+2}$, so bus 0 gets 3, bus 1 gets 5, bus 2 gets 7, and so on, exactly as you said.

The assignment. Old guest currently in room n ($n = 1, 2, 3, \dots$) moves to room 2^n . New guest in seat s ($s = 1, 2, 3, \dots$) of bus m moves to room q_m^s .

Proposition. No two guests, old or new, are ever sent to the same room.

Proof. Every room number produced here is a single prime raised to a positive power, so by the Fundamental Theorem of Arithmetic its entire factorization is that one prime, to that one exponent, nothing else. Two old guests: $2^n = 2^{n'}$ forces $n = n'$. Two new guests on the *same* bus: $q_m^s = q_m^{s'}$ forces $s = s'$, since a fixed base > 1 raised to different powers gives different values. A new guest on bus m against a new guest on a different bus m' : $q_m^s = q_{m'}^{s'}$ is impossible since $q_m \neq q_{m'}$ are different primes, so the two sides would have different prime factorizations unless both exponents are 0, which never happens here ($s, s' \geq 1$). An old guest against any new guest: $2^n = q_m^s$ is impossible the same way, since q_m is always an odd prime, never 2. Every case rules out a collision, so the assignment is injective. ■

That single fact, that we never run out of distinct primes, is doing all the work: it's the same unique-prime-factorization trick as page 2, just handed out one prime per group instead of only two primes total.

Still not surjective

Room 1 is never used (2^0 and q_m^0 are never assigned, since $n, s \geq 1$). Room $6 = 2 \times 3$ is never used either: it has *two* distinct prime factors, but every room this scheme produces has exactly *one*. The same goes for $10, 12, 14, 15, \dots$, anything that isn't a pure prime power is permanently skipped. So, once again: injective, solves the hotel, but not surjective, so not a bijection, and not by itself a proof that this whole infinite collection of guests has exactly the same size as $\mathbb{N}$.

What this construction actually proves, once you add the usual "injects into a countable set $\Rightarrow$ countable" step, is the union version of our theorems: countably many buses, each countably infinite, plus the original countably infinite hotel, all combined, is still just countably infinite. That's Act 2 (one extra bus) generalized all the way up to infinitely many buses at once.